AAE5210

Space Vehicle Propulsion Systems

2026/27 Semester 1

Instructor: Yiqing WANG

Miniproject 1

Low Thrust Orbit Simulation

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Code, figures and results

https://github.com/xuytwinter/AAE5210_Miniproject1

MiNi Project 1

1. Nondimensionalization

Let $\mu = GM$, and define: $r = \frac{R}{R_0}$, $\tau = \frac{t}{T_0}$, $T_0 = \sqrt{\frac{R_0^3}{\mu}}$.

Dots denote derivatives with respect to t, and primes with respect to τ. Thus:

$$\dot{R} = \frac{R_0}{T_0} r', \quad \ddot{R} = \frac{R_0}{T_0^2} r'', \quad \dot{\theta} = \frac{\theta'}{T_0}, \quad \ddot{\theta} = \frac{\theta''}{T_0^2}.$$

Substituting into the radial equation gives:

$$\frac{R_0}{T_0^2}(r'' - r\theta'^2) = -\frac{\mu}{R_0^2 r^2}.$$

Dividing by $\frac{R_0}{T_0^2}$ and using $\frac{\mu T_0^2}{R_0^3} = 1$, we obtain:

$$r'' - r\theta'^2 = -\frac{1}{r^2}$$

Substituting $a_\theta = \frac{0.005/\mu}{R_0^2}$ into the transverse equation gives:

$$\frac{R_0}{T_0^2}(r\theta'' + 2r'\theta') = 0.005\frac{\mu}{R_0^2}.$$

$$r\theta'' + 2r'\theta' = 0.005$$

The initial angular velocity becomes:

$$\theta'(0) = T_0\sqrt{\frac{\mu}{R^3}} = 1$$

Thus: $r(0) = 1$, $r'(0) = 0$, $\theta(0) = 0$, $\theta'(0) = 1$

1. Nondimensionalization (continued)

Scaling the specific mechanical energy by $\frac{\mu}{R_0}$ gives:

$$\mathcal{E} = \frac{\frac{1}{2}\left[\dot{R}^2 + (R\dot{\theta})^2\right] - \frac{\mu}{R}}{\frac{\mu}{R_0}}, \quad \mathcal{E} = \frac{1}{2}\left[r'^2 + (r\theta')^2\right] - \frac{1}{r}.$$

$$\mathcal{E}(\theta) = -\frac{1}{2}.$$

Miniproject 1

2. Numerical method and results.

① First-order system

Let $u = r'$ and $w = \theta'$. Thus:

$$r' = u,\quad u' = rw^2 - \frac{1}{r^2},\quad \theta' = w,\quad w' = \frac{-0.005 - 2uw}{r}.$$

Initial state is $(r, u, \theta, w) = (1, 0, 0, 1)$. So we can use Python solve_ivp with DoP853, relative tolerance 10^{-11}, absolute tolerance 10^{-13}, and max step 0.2.

② 8 positive-axis crossing

Initial position counts as crossing 1. Thus, crossing 8 occur at:

$$\theta = 2\pi(8-1) = 14\pi.$$

The upward zero of $g_8 = \theta - 14\pi$ is recorded.

③ Escape time

$$\mathcal{E} = \frac{1}{2}\left[u^2 + (rw)^2\right] - \frac{1}{r}.$$

Integration stops when $\mathcal{E}$ first reaches zero from below.

④ Result and pictures.

will show in pdf.

3. Baseline results: trajectory

The model assumes constant mass, planar central gravity and continuous transverse acceleration with $\alpha = 0.005$. The initial position counts as crossing 1.

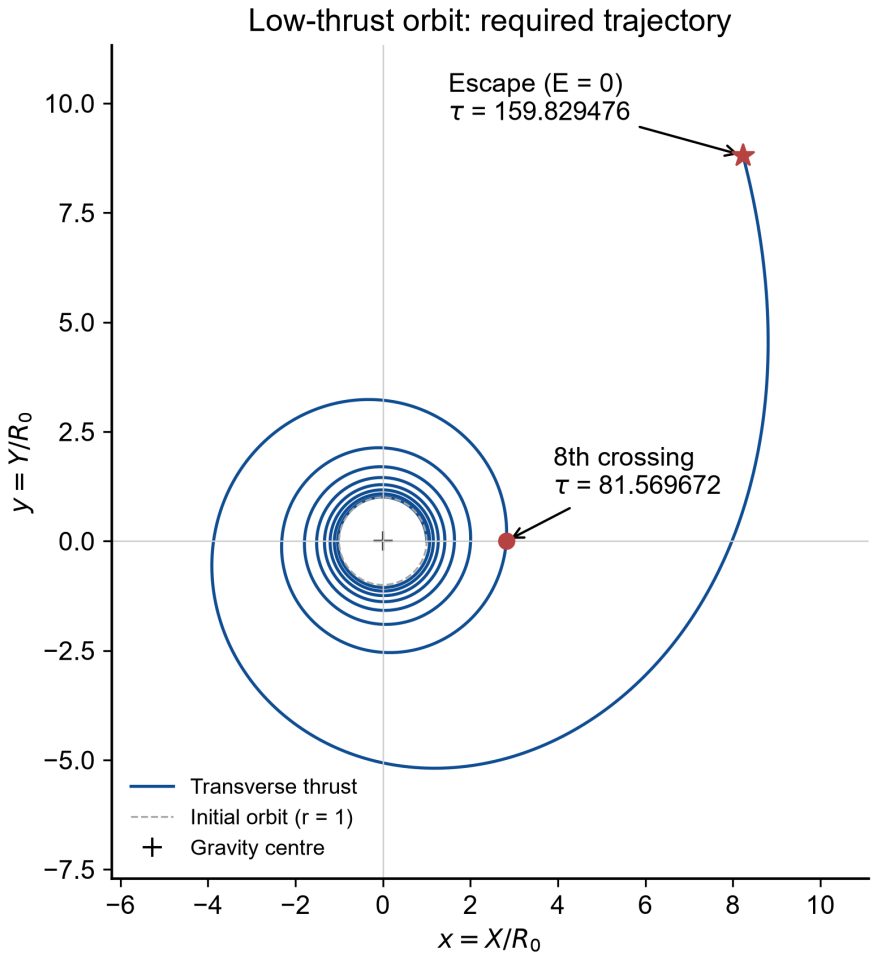


Figure 1. Dimensionless trajectory, with the eighth positive-axis crossing and first zero-energy point marked. Axes have equal scale; the dashed circle is the initial orbit.

Event	Time (dimensionless)	Position (x, y)
8th positive-axis crossing	81.569672	(2.821324, 0)
First zero-energy event	159.829476	(8.230888, 8.808885)

Thus $t_8 = 81.569672 T_0$ and $t_{\text{esc}} = 159.829476 T_0$.

3. Baseline results: energy and validation

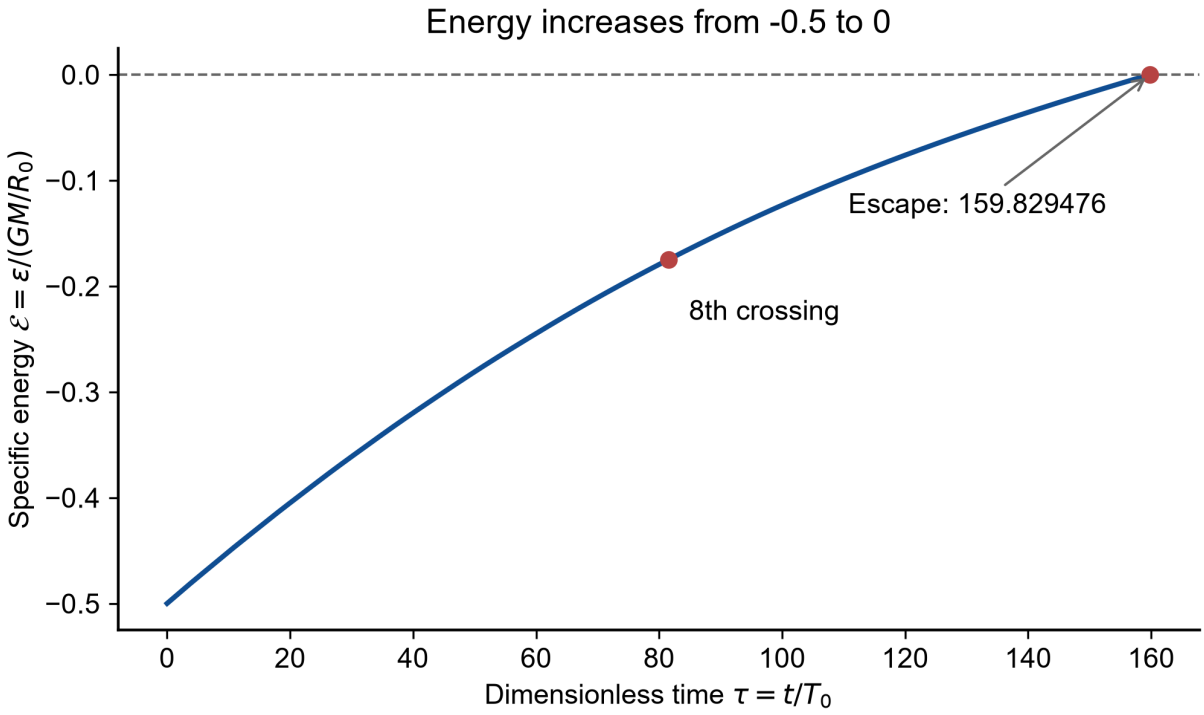


Figure 2. Specific mechanical energy increases from -0.5 to zero. The eighth crossing occurs while $\mathcal{E} = -0.174767313$, so escape has not yet occurred.

At escape, $r = 12.055869129$ and $v = 0.407301243$. These values satisfy $v = \sqrt{2/r}$, consistent with zero specific energy.

Numerical convergence

Method	Relative tolerance	Max. step	8th crossing	Escape
DOP853	1e-08	1.0	81.569672	159.829476
DOP853	1e-10	0.5	81.569672	159.829476
DOP853	1e-12	0.1	81.569672	159.829476
RK45	1e-11	0.1	81.569672	159.829476

The baseline DOP853 run uses relative tolerance 10^{-11}, absolute tolerance 10^{-13}, and maximum step 0.2. Both event times remain unchanged to six decimal places in the table. Independent Cartesian integration also agrees to this precision.

Thrust does work and applies torque: $\mathcal{E}' = \alpha r\omega$ and $(r^2\omega)' = \alpha r$. The integrated work and torque balances have maximum residuals of approximately 3.03×10^{-15} and 9.77×10^{-15}.

4. Extension A: comparison of thrust directions

The alternative model keeps the same thrust magnitude but directs thrust along the instantaneous velocity. All other assumptions are unchanged.

$$v = \sqrt{u^2 + (r\omega)^2}, \qquad a_r = \alpha\frac{u}{v}, \qquad a_\theta = \alpha\frac{r\omega}{v}.$$

$$r' = u, \quad u' = r\omega^2 - \frac{1}{r^2} + a_r, \quad \theta' = \omega, \quad \omega' = \frac{a_\theta - 2u\omega}{r}.$$

The transverse model instead uses $a_r = 0$, $a_\theta = \alpha$. The mission objective is escape-time prediction. Taking velocity-aligned thrust as the reference, a 1% relative-error tolerance is selected:

$$\delta_t = \frac{|\tau_{\text{esc, T}} - \tau_{\text{esc, V}}|}{\tau_{\text{esc, V}}} \le 0.01.$$

The sweep uses $\alpha \in \{0.001, 0.0025, 0.005, 0.01, 0.02\}$. All cases start at $r_0 = 1$, $\theta_0 = 0$. The five initial pairs (u_0, ω_0) are circular: $(0, 1)$; outward: $(0.2, 1)$; inward: $(-0.2, 1)$; slow: $(0, 0.9)$; and fast: $(0, 1.1)$. All are initially bound.

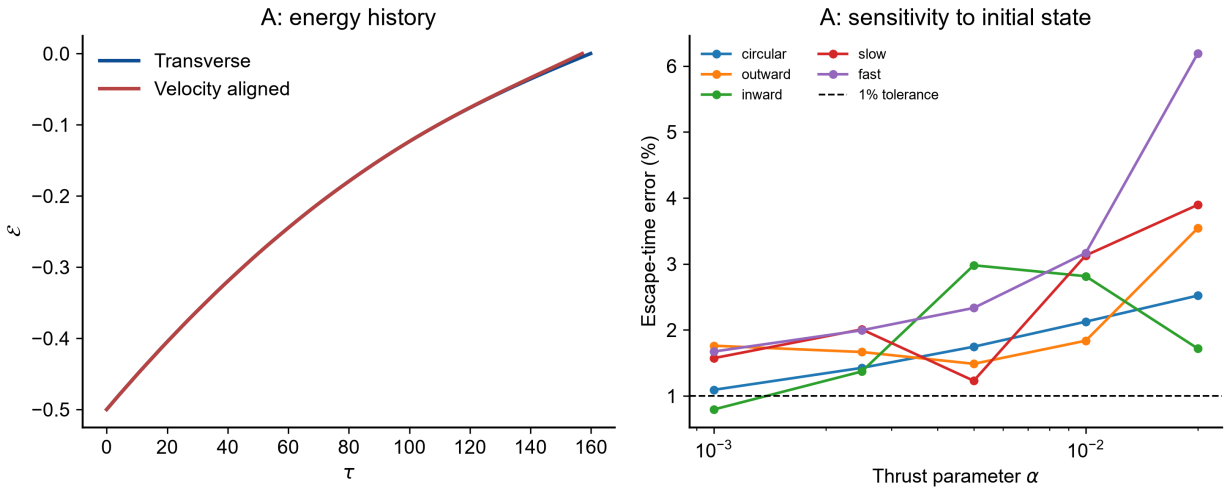


Figure 3. Left: baseline energy histories, each ending at its own escape time. Right: escape-time error for 25 paired cases; the dashed line is the selected 1% tolerance.

For the prescribed circular case, transverse thrust gives 159.829476, whereas velocity-aligned thrust gives 157.085970. The relative difference is 1.7465%, so the transverse approximation fails the selected 1% objective (but satisfies a 2% objective for this case).

4. Extension A: findings and reproducibility

Effect of thrust magnitude and initial state

Initial state	0.001	0.0025	0.005	0.01	0.02
circular	1.093%	1.426%	1.746%	2.126%	2.521%
outward	1.761%	1.666%	1.487%	1.837%	3.542%
inward	0.795%	1.371%	2.980%	2.815%	1.720%
slow	1.573%	2.009%	1.230%	3.129%	3.896%
fast	1.673%	1.995%	2.334%	3.167%	6.189%

Table. Relative escape-time errors, with columns giving the thrust parameter. Only the inward case at 0.001 meets the 1% criterion among the sampled cases.

For circular initial conditions, the sampled error increases with thrust magnitude. Noncircular initial states show nonmonotonic behaviour, so thrust magnitude alone is not a sufficient acceptance criterion. The table represents discrete cases, not a continuous boundary of validity.

At a common instantaneous state, the energy rates are $\alpha r\omega$ for transverse thrust and αv for velocity-aligned thrust. The directions are locally close when $|u| \ll |r\omega|$. Radial motion becomes important near escape, explaining why an initially circular, low-thrust transfer need not remain within 1% escape-time agreement.

Conclusion

For the prescribed model, the eighth crossing occurs at 81.569672 and escape at 159.829476 in dimensionless time. The two thrust directions should not be interchanged for a 1% escape-time objective in the baseline case.

Reproducible program

The accompanying solve.py reproduces the event times, all three figures, the convergence study and the 25 paired extension cases. Install NumPy, SciPy and Matplotlib, then run: python solve.py.

The solver uses continuous event interpolation. It stores the unwrapped angle, records the eighth-crossing event without stopping, and terminates at the first upward zero of energy. The work and torque diagnostic variables do not alter the four dynamical states.

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